# Hazem Ilahi

**Data Science & AI Engineering Student — Machine Learning | MLOps | LLM / RAG**

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**Seeking 6-month end-of-studies internship (PFE), from Jan/Feb 2027 — open to relocation**

*Mandatory curricular internship — tripartite internship agreement signed by ESPRIT*

## PROFILE

Final-year engineering student in Data Science & AI at ESPRIT (graduating 2027) with hands-on experience in machine learning, MLOps and LLM/RAG systems. During my last internship I designed the ML detection layer of a digital-forensics framework: anomaly detection plus a calibrated gradient-boosting classifier that raised attack-stage coverage from 18% to 84% at 1% false positives, shipped with a weekly retraining pipeline (champion/challenger, drift monitoring, automatic rollback). Looking for an applied data science / ML engineering internship: anomaly detection, predictive analytics, LLM/RAG applications, MLOps, forecasting.

## EDUCATION

**ESPRIT — École Supérieure Privée d'Ingénierie et de Technologies** 2023 – 2027 (expected)

*Engineering degree program, Data Science & Artificial Intelligence — final year; 6-month mandatory end-of-studies internship (PFE) from Jan/Feb 2027* *Ariana, Tunisia*

**IPEIT — Preparatory Institute for Engineering Studies of Tunis** 2020 – 2023

*Two-year preparatory program for national engineering-school entrance exams, Physics–Chemistry* *Montfleury, Tunis, Tunisia*

## EXPERIENCE

**Machine Learning Intern — NEXT STEP IT (Security Operations Center)** Jul 2026 – Sep 2026

*ATOR DFIR — ML threat-detection layer and MLOps pipeline* *Charguia 1, Tunisia*

- Raised attack-stage coverage on a public adversary-emulation corpus from 18% (Sigma rules alone) to 84% (rules + ML) at a 1% false-positive rate, recovering 135 of the 168 stages missed by rules.
- Designed and integrated the ML layer of ATOR DFIR, a forensics and threat-hunting framework for Windows/Linux endpoints, on top of the security team's YARA/Sigma rule engine.
- Threat-likelihood classifier (histogram gradient boosting, Platt calibration): PR-AUC 0.927 vs 0.278 for rules, 1.6% calibration error, 5-fold cross-validation grouped by capture.
- Unsupervised anomaly detection (Isolation Forest, 300 trees): 23 of the top 25 flags on the benchmark were true attack stages; MITRE ATT&CK tactic classifier with confidence threshold (79.7% precision).
- Built the training set from 115 public attack captures (Empire, Covenant, Mimikatz, Metasploit) via ETL into the framework schema, labeled by process lineage; 110 behavioral features per process, computed by one code path shared by training and production.
- Weekly MLOps pipeline: snapshot, ETL, retraining, 14 offline quality gates, 7-day silent champion/challenger trial on live traffic, atomic deployment with canary check, automatic rollback, drift monitoring (PSI) and protection of retraining against reference poisoning.
- Integration: 10 REST endpoints and a threat-hunting page for SOC analysts (P1–P4 priorities, verdicts); 500+ automated tests, including fault-injection tests guaranteeing that an ML failure never interrupts rule-based detection.

*Stack: Python 3.13, scikit-learn, pandas, NumPy, SQLite, FastAPI, Jinja2, pytest, Git*

**Software Developer Intern — NEXT STEP IT** Jul 2025 – Sep 2025

*SOC report-generation web platform (pair internship)* *Charguia 1, Tunisia*

- Co-designed and developed, in a pair, a Python/Django web platform for SOC analysts with an intuitive UI/UX.
- Automated SOC report generation, replacing repetitive manual writing; prepared the solution for future deployment.

*Stack: Python, Django, HTML/CSS, Git*

## PROJECTS

**TurboWatch — Predictive maintenance copilot for turbofan engines** 2026

*Personal project — live demo: dyaus2.github.io/turbowatch · code: github.com/Dyaus2/turbowatch*

- Remaining-useful-life prediction on NASA C-MAPSS turbofan data: gradient boosting with a P10–P90 band and a 1D-CNN (PyTorch) on causal, regime-normalized features shared by training and serving; test RUL RMSE 15.5 cycles (CNN) / 16.3 (GBM), mean of 5 seeds, engine-grouped splits.
- Calibrated risk and a k-of-n alarm policy tuned on a cost matrix: 97% of test-set failures detected a median 36 cycles ahead at a 2% false-alarm rate; SHAP explanation for every prediction.
- RAG maintenance copilot over FAA/NASA documents (bge-small + BM25, RRF, cross-encoder reranking): cited answers, refusal when out of scope; correct passage ranked first on 26/26 test questions, also when running fully in the browser (transformers.js).
- MLOps and delivery: MLflow, PSI drift detection with champion/challenger retraining (FD003 test RMSE 20.7 → 15.0), FastAPI + React/TypeScript dashboard with live replay, Docker, GitHub Actions CI.

*Stack: Python, scikit-learn, PyTorch, SHAP, MLflow, FastAPI, sentence-transformers, React, TypeScript, Docker*

**UnderHood (Garage Augmenté) — Multi-agent RAG assistant for automotive pre-diagnosis** 2026 – 2027

*ESPRIT team project — owner of the knowledge base and retrieval layer*

- Corpus: 7 resumable Python collectors gathering 2,019 owner's manuals (≈697,000 pages, 26 brands, 73% official sources), covering 54 of the 56 priority models of the Tunisian market. SHA-256 deduplication, Tesseract OCR (≈18,500 pages), 364 model families, Qdrant index of 101,417 passages.

- Retrieval: query expansion in French / English / Tunisian Derja plus LLM query rewriting, hybrid BGE-M3 + BM25 search (RRF), brand/model/year filtering, cross-encoder reranking and exact lookup of 9,533 OBD-II codes, all exposed as a FastAPI service to the diagnostic agent (LangGraph).
- Evaluation: ablation on 40 questions (hit@5, MRR, double-blind LLM judging): the vehicle filter raises the share of correct answers found in the right manual from 0% to 72% (MRR 0.63).

*Stack: Python, Qdrant, BGE-M3, BM25, FastAPI, LangGraph, PyMuPDF, Tesseract*

**DressMe — Smart personal fashion assistant for the Tunisian market** 2026 – 2027

*ESPRIT team project (7 people, Advanced Data Science Project) — from design to working prototype (API + mobile app)*

- Style representation: benchmarked encoders on Polyvore (152,785 items, 95% bootstrap CIs). Chose Qwen3-VL-Embedding-2B, as accurate as the 8B model but 3.7× faster. "Fashion DNA" learned by a Set Transformer trained contrastively (InfoNCE).
- Personalized outfits: compatibility head per category pair (AUC 0.852 vs 0.830 for cosine alone). Beam-search generation conditioned on weather and occasion. Personalization by Thompson sampling with a population prior, evaluated on 40 simulated personas over 60 days.
- Vision and generative AI: SAM 3 segmentation and garment recognition from photos (96.6% accuracy in auto-fill). Diffusion-based virtual try-on (Qwen Image Edit via ComfyUI). Local LLM stylist (Qwen 9B) with every sentence checked against facts: 199 of 200 answers grounded.
- Explainable AI and marketplace: exact per-criterion contributions to the score; occlusion maps evaluated for faithfulness (deletion/insertion curves); concepts shown only above 0.85 AUC; second-hand prices predicted by quantile LightGBM with conformal calibration (80.0% coverage on 423k listings), explained with TreeSHAP.

*Stack: Python, PyTorch, FastAPI, LightGBM, llama.cpp, React Native (Expo)*

**Kadastra — AI real-estate intelligence platform** 2026

*ESPRIT team project (Data Crunchers) — Microsoft TDSP methodology*

- End-to-end pipeline for the Tunisian real-estate market: collection of ≈26,000 listings (Mubawab, Tayara), XGBoost price estimation, contract and investment-scenario generation.
- Removed a data leak in the model features; calibrated financial constants on Tunisian legislation and Central Bank (BCT) rates; full-stack deployment with FastAPI, Django, React and Docker.

**Show, Attend and Tell — Image captioning (computer vision + NLP)** 2026

*Deep learning project — PyTorch, Flickr8k*

- Reproduced the CNN + LSTM attention encoder-decoder, then extended it with a ViT encoder, CLIP reranking of beam search and BLIP fine-tuning; evaluated with BLEU and attention-map visualization.

**Time-series forecasting — U.S. construction spending** 2026

*Statistical modeling project in R*

- Three-phase analysis of residential construction spending (U.S. Census Bureau): graphical analysis, deterministic polynomial trends, ARIMA/SARIMA modeling with diagnostics and forecasts.

## TECHNICAL SKILLS

**Programming:** Python, R, SQL, C/C++, Java, PHP
**ML / DL:** scikit-learn, XGBoost, PyTorch, TensorFlow, Keras; anomaly detection, classification, probability calibration, grouped cross-validation; computer vision (CNN, ViT, CLIP, BLIP); time-series forecasting (ARIMA/SARIMA)
**LLM / RAG / NLP:** RAG pipelines, Qdrant, BGE-M3 embeddings, BM25, hybrid search, cross-encoder reranking, retrieval evaluation (hit@k, MRR), Transformers
**MLOps:** automated retraining, model registry, MLflow, champion/challenger deployment, drift monitoring (PSI), quality gates, canary deployment and rollback, pytest
**Data:** pandas, NumPy, ETL, web scraping, OCR (Tesseract), SQLite, Power BI, Tableau
**Cloud & deployment:** AWS (Cloud Foundations), Docker, Docker Compose, FastAPI, Django, Flask, Spring Boot, Symfony, REST APIs, n8n, Git
**Methods:** Microsoft TDSP, CRISP-DM, Design Thinking, Agile/Scrum

## CERTIFICATIONS

- AWS Cloud Foundations — Amazon Web Services (badge, Nov 2026) — verify on Credly
- Fundamentals of Deep Learning — NVIDIA (2026)
- Applications of AI for Anomaly Detection — NVIDIA (2026)
- Building Transformer-Based Natural Language Processing Applications — NVIDIA (2026)
- CCNA 2: Switching, Routing, and Wireless Essentials (SRWE) — Cisco Networking Academy (2025)

## LANGUAGES

English: fluent | French: advanced (professional working proficiency) | Arabic: native